Middle East Technical University
Department of Electrical & Electronics Engineering

EE 533 – Information Theory

Assignment 1 – Solutions

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Problems: 2.2, 2.3, 2.10, 2.39, 3.3
Textbook: Cover & Thomas, *Elements of Information Theory*, 2nd ed. (Wiley, 2006)

All logarithms are base 2 (bits) unless otherwise noted.

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1. Problem 2.2 – Entropy of Functions

Problem Statement

Let X be a random variable taking on a *finite* number of values. What is the (general) inequality relation between $\mathrm{H}(X)$ and $\mathrm{H}(Y)$ if:

(a) $Y = 2^X$

(b) $Y = \cos X$

Background

We recall two fundamental results from lecture (Week 1).

Theorem 1.1 (Invariance under relabelling)**.** *For any one-to-one (injective) function f,*

$$\mathrm{H}(f(X)) = \mathrm{H}(X).$$

Theorem 1.2 (Data-processing / Function inequality)**.** *For any (possibly non-injective) function g,*

$$\mathrm{H}(g(X)) \;\leq\; \mathrm{H}(X),$$

with equality if and only if g is injective on $\mathrm{supp}(p_X) := \{x : p_X(x) > 0\}$.

Solution

(a) $Y = 2^X$

Step 1 – Show $f(x) = 2^x$ is injective. Suppose $2^{x_1} = 2^{x_2}$ for some x_1, x_2 in the support of p_X. Taking $\log_2$ of both sides (which is the inverse of 2^x, hence well-defined since $2^x > 0$ for all x):

$$2^{x_1} = 2^{x_2} \implies \log_2(2^{x_1}) = \log_2(2^{x_2}) \implies x_1 = x_2.$$

Hence $f(x) = 2^x$ is **injective** on any set of real numbers.

Step 2 – Apply Theorem 1.1. Because f is injective, Theorem 1.1 gives immediately:

$$\boxed{\mathrm{H}(Y) = \mathrm{H}(X).}$$

Note. Knowing $Y = 2^X$ is *exactly* as informative as knowing X: given Y one can recover $X = \log_2 Y$ uniquely. No information is created or destroyed by a bijective relabelling.

(b) $Y = \cos X$

Step 1 – Cosine is not injective in general. The cosine function satisfies $\cos(\theta) = \cos(-\theta)$ for all θ. Therefore if $\mathrm{supp}(p_X)$ contains both θ and $-\theta$ for some $\theta \neq 0$, then $g(x) = \cos(x)$ is not injective on $\mathrm{supp}(p_X)$, and *information is lost* in the mapping $X \mapsto Y$.

Step 2 – Formal chain-rule argument. Since $Y = g(X)$ is a deterministic function of X, we have $H(Y \mid X) = 0$, and hence $H(X, Y) = H(X)$. By the chain rule:

$$\mathrm{H}(X) \;=\; \mathrm{H}(X, Y) \;=\; \mathrm{H}(Y) + \mathrm{H}(X \mid Y).$$

Since $\mathrm{H}(X \mid Y) \geq 0$, it follows that

$$\boxed{\mathrm{H}(Y) \;\leq\; \mathrm{H}(X).}$$

Step 3 – Equality condition. The proof shows:

$$\mathrm{H}(Y) = \mathrm{H}(X) \iff \mathrm{H}(X \mid Y) = 0 \iff X \text{ is a (deterministic) function of } Y \iff g \text{ is injective on}$$

Example where equality holds. If $\mathrm{supp}(p_X) \subseteq [0, \pi]$, then cosine is strictly decreasing (hence injective) on that interval, so $\mathrm{H}(\cos X) = \mathrm{H}(X)$. The inequality is therefore strict only when g causes genuine "collisions" in the support.

Notation. Throughout this problem we use $f(x) = 2^x$ for part (a) and $g(x) = \cos x$ for part (b) to keep the two functions notationally distinct.

2. Problem 2.3 – Minimum Entropy

Problem Statement

What is the minimum value of

$$\mathrm{H}(p_1, \ldots, p_n) \;=\; \mathrm{H}(\mathbf{p})$$

as $\mathbf{p}$ ranges over the set of all n-dimensional probability vectors? Find all $\mathbf{p}$'s that achieve this minimum.

Background

Definition 2.1 (Shannon Entropy)**.** For a discrete random variable X with pmf p_X,

$$\mathrm{H}(X) \;:=\; -\sum_x p_X(x)\,\log p_X(x) \;=\; \mathbf{E}\left[\log \frac{1}{p_X(X)}\right],$$

where we adopt the convention $0 \log 0 \;:=\; 0$. *Motivation of the convention:* By L'Hôpital's rule, $\lim_{p\to 0^+} p \log p = 0$, so the convention preserves continuity of each summand at $p_i = 0$.

Solution

Theorem 2.2.

$$\min_{\mathbf{p}\in\Delta_n} \mathrm{H}(\mathbf{p}) \;=\; 0,$$

where $\Delta_n := \{\mathbf{p} \in [0,1]^n : \sum_{i=1}^n p_i = 1\}$. *The minimum is achieved* if and only if $\mathbf{p} = \mathbf{e}_k$ *for some* $k \in \{1, \ldots, n\}$ *(a point-mass / deterministic distribution).*

Proof. We proceed in three steps.

Step 1 – Lower bound $\mathrm{H}(\mathbf{p}) \geq 0$.

For every $p_i \in [0,1]$ we have $\log p_i \leq 0$, so $-p_i \log p_i \geq 0$ (with the convention $0 \cdot \log 0 = 0$). Since the entropy is a sum of non-negative terms,

$$\mathrm{H}(\mathbf{p}) = -\sum_{i=1}^n p_i \log p_i \;\geq\; 0.$$

Step 2 – Achievability.

Set $p_k = 1$ for some fixed k and $p_i = 0$ for all $i \neq k$. Then

$$\mathrm{H}(\mathbf{p}) = -1 \cdot \log 1 - \sum_{i\neq k} 0 \cdot \log 0 = 0 - 0 = 0.$$

Step 3 – Uniqueness.

Suppose $\mathrm{H}(\mathbf{p}) = 0$. Then every summand $-p_i \log p_i = 0$. Since $-p \log p = 0$ if and only if $p \in \{0,1\}$ (the function $-p \log p$ is strictly positive for $p \in (0,1)$), we conclude $p_i \in \{0,1\}$ for all i. Combined with $\sum_{i=1}^n p_i = 1$, exactly one $p_k = 1$ and all others equal 0. There are exactly n such vectors: $\mathbf{e}_1, \ldots, \mathbf{e}_n$. □

$$\boxed{\min_{\mathbf{p}\in\Delta_n} \mathrm{H}(\mathbf{p}) \;=\; 0,}$$

achieved **if and only if** $\mathbf{p} \in \{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$, i.e., the distribution is a point mass (the outcome is *deterministic*).

Note. The maximum is $\mathrm{H}(\mathbf{p}) \leq \log n$, attained uniquely at the uniform distribution $p_i = 1/n$ for all i. This follows from log-sum inequality / Jensen's inequality applied to the strictly concave function $-p \log p$.

3. Problem 2.10 – Entropy of a Disjoint Mixture

Problem Statement

Let X_1 and X_2 be discrete random variables drawn according to pmfs $p_1(\cdot)$ and $p_2(\cdot)$ over *disjoint* alphabets $\mathcal{X}_1 = \{1, \ldots, m\}$ and $\mathcal{X}_2 = \{m+1, \ldots, n\}$. Define the mixture variable

$$X = \begin{cases} X_1 & \text{with probability } \alpha, \\ X_2 & \text{with probability } 1-\alpha, \end{cases} \qquad \alpha \in (0,1).$$

(a) Find $\mathrm{H}(X)$ in terms of $\mathrm{H}(X_1)$, $\mathrm{H}(X_2)$, and α.

(b) Maximize over α to show $2^{\mathrm{H}(X)} \le 2^{\mathrm{H}(X_1)} + 2^{\mathrm{H}(X_2)}$ and interpret using the notion that $2^{\mathrm{H}(X)}$ is the effective alphabet size.

Background

Denote the *binary entropy function* $\mathrm{h}(\alpha) := -\alpha \log \alpha - (1-\alpha)\log(1-\alpha)$ for $\alpha \in (0,1)$.

Solution

(a) Expression for $\mathrm{H}(X)$

Step 1 – PMF of X. Because $\mathcal{X}_1 \cap \mathcal{X}_2 = \emptyset$, the pmf of X is:

$$p_X(k) = \begin{cases} \alpha\, p_1(k) & k \in \mathcal{X}_1, \\ (1-\alpha)\, p_2(k) & k \in \mathcal{X}_2. \end{cases}$$

Step 2 – Split the entropy sum.

$$\mathrm{H}(X) = -\sum_{k\in\mathcal{X}_1} \alpha p_1(k) \log\Big(\alpha p_1(k)\Big) - \sum_{k\in\mathcal{X}_2} (1-\alpha) p_2(k) \log\Big((1-\alpha) p_2(k)\Big).$$

Step 3 – Expand $\log(\alpha p_i) = \log\alpha + \log p_i$.

$$\begin{aligned}
\mathrm{H}(X) &= -\sum_{k\in\mathcal{X}_1} \alpha p_1(k)\Big[\log\alpha + \log p_1(k)\Big] - \sum_{k\in\mathcal{X}_2} (1-\alpha) p_2(k)\Big[\log(1-\alpha) + \log p_2(k)\Big] \\
&= \underbrace{-\alpha\log\alpha}\underbrace{\sum_k p_1(k)}_{=1} + \alpha\underbrace{\Big(-\sum_k p_1(k)\log p_1(k)\Big)}_{=\mathrm{H}(X_1)} \\
&\quad - \underbrace{(1-\alpha)\log(1-\alpha)}\underbrace{\sum_k p_2(k)}_{=1} + (1-\alpha)\underbrace{\Big(-\sum_k p_2(k)\log p_2(k)\Big)}_{=\mathrm{H}(X_2)}.
\end{aligned}$$

Step 4 – Collect terms.

$$\boxed{\mathrm{H}(X) \;=\; \mathrm{h}(\alpha) + \alpha\,\mathrm{H}(X_1) + (1-\alpha)\,\mathrm{H}(X_2).}$$

Interpretation. The entropy of the mixture decomposes into two parts:

1. $\mathrm{h}(\alpha)$: the uncertainty about *which component* was selected.
2. $\alpha \mathrm{H}(X_1) + (1-\alpha)\mathrm{H}(X_2)$: the expected uncertainty *within* the selected component.

This is the chain rule in disguise: $\mathrm{H}(X) = \mathrm{H}(B) + \mathrm{H}(X \mid B)$, where B is the Bernoulli(α) selector.

(b) Upper bound $2^{\mathrm{H}(X)} \leq 2^{\mathrm{H}(X_1)} + 2^{\mathrm{H}(X_2)}$

Step 1 – Rewrite in terms of a and b. Let $a := 2^{\mathrm{H}(X_1)}$ and $b := 2^{\mathrm{H}(X_2)}$. Using the result of part (a):

$$\begin{aligned}
2^{\mathrm{H}(X)} &= 2^{\,\mathrm{h}(\alpha) + \alpha\,\mathrm{H}(X_1) + (1-\alpha)\,\mathrm{H}(X_2)} \\
&= 2^{\mathrm{h}(\alpha)} \cdot 2^{\,\alpha\,\mathrm{H}(X_1)} \cdot 2^{\,(1-\alpha)\,\mathrm{H}(X_2)} \\
&= 2^{\mathrm{h}(\alpha)} \cdot \left(2^{\mathrm{H}(X_1)}\right)^{\alpha} \cdot \left(2^{\mathrm{H}(X_2)}\right)^{1-\alpha} \\
&= 2^{\mathrm{h}(\alpha)} \cdot a^{\alpha} \cdot b^{1-\alpha}.
\end{aligned}$$

Step 2 – Maximize $\mathrm{H}(X)$ over α. Differentiating with respect to α and setting equal to zero:

$$\begin{aligned}
\frac{d}{d\alpha}\mathrm{H}(X) &= \frac{d}{d\alpha}\Big[\mathrm{h}(\alpha) + \alpha \log a + (1-\alpha)\log b\Big] \\
&= \frac{d\mathrm{h}}{d\alpha} + \log a - \log b \\
&= \log\frac{1-\alpha}{\alpha} + \log a - \log b \;=\; 0,
\end{aligned}$$

where we used $\mathrm{h}'(\alpha) = \log\frac{1-\alpha}{\alpha}$. Solving:

$$\log\frac{1-\alpha}{\alpha} = \log\frac{b}{a} \implies \frac{1-\alpha}{\alpha} = \frac{b}{a} \implies \boxed{\alpha^* = \frac{a}{a+b} = \frac{2^{\mathrm{H}(X_1)}}{2^{\mathrm{H}(X_1)} + 2^{\mathrm{H}(X_2)}}.}$$

Since $\mathrm{H}(X)$ is strictly concave in α (as a sum of concave functions), this critical point is the *global maximum*.

Step 3 – Evaluate $2^{\mathrm{H}(X)}$ at α^*. At α^*:

$$\mathrm{h}(\alpha^*) = -\frac{a}{a+b}\log\frac{a}{a+b} - \frac{b}{a+b}\log\frac{b}{a+b} = \frac{a}{a+b}\log\frac{a+b}{a} + \frac{b}{a+b}\log\frac{a+b}{b}.$$

Therefore:

$$\begin{aligned}
2^{\mathrm{H}(X^*)} &= 2^{\mathrm{h}(\alpha^*)} \cdot a^{\frac{a}{a+b}} \cdot b^{\frac{b}{a+b}} \\
&= \left(\frac{a+b}{a}\right)^{\frac{a}{a+b}} \cdot \left(\frac{a+b}{b}\right)^{\frac{b}{a+b}} \cdot a^{\frac{a}{a+b}} \cdot b^{\frac{b}{a+b}} \\
&= (a+b)^{\frac{a}{a+b}} \cdot (a+b)^{\frac{b}{a+b}} \;=\; (a+b)^{\frac{a+b}{a+b}} \;=\; a+b.
\end{aligned}$$

Step 4 – Conclude the bound. Since α^* is the *maximizer* of $\mathrm{H}(X)$, for *every* $\alpha \in (0,1)$:

$$\mathrm{H}(X) \;\le\; \mathrm{H}(X^*), \quad \text{hence} \quad 2^{\mathrm{H}(X)} \;\le\; 2^{\mathrm{H}(X^*)}.$$

$$\boxed{2^{\mathrm{H}(X)} \;\le\; 2^{\mathrm{H}(X_1)} + 2^{\mathrm{H}(X_2)},}$$

with equality if and only if $\alpha = \alpha^* = \dfrac{2^{\mathrm{H}(X_1)}}{2^{\mathrm{H}(X_1)} + 2^{\mathrm{H}(X_2)}}$.

Interpretation. The quantity $2^{\mathrm{H}(X)}$ represents the *effective alphabet size*: the number of roughly equally likely outcomes. The bound states that the effective alphabet size of a mixture is at most the *sum* of the effective alphabet sizes of the components. The optimal mixing weight α^* is proportional to the effective size of each component, balancing the two distributions to maximize total uncertainty.

4. Problem 2.39 – Entropy and Pairwise Independence

Problem Statement

Let X, Y, Z be three binary Bernoulli($\frac{1}{2}$) random variables that are *pairwise independent*:

$$\mathrm{I}(X;Y) = \mathrm{I}(X;Z) = \mathrm{I}(Y;Z) = 0.$$

(a) Under this constraint, what is the *minimum* value of $\mathrm{H}(X,Y,Z)$?

(b) Give an example achieving this minimum.

Background

Definition 4.1 (Mutual Information)**.** $\mathrm{I}(X;Y) := \mathrm{H}(X) + \mathrm{H}(Y) - \mathrm{H}(X,Y)$. Hence $\mathrm{I}(X;Y) = 0 \iff \mathrm{H}(X,Y) = \mathrm{H}(X) + \mathrm{H}(Y)$ (i.e., X and Y are *independent*).

Solution

(a) Minimum value of $\mathrm{H}(X,Y,Z)$

Step 1 – Marginal entropies. Since each variable is Bernoulli($\frac{1}{2}$):

$$\mathrm{H}(X) = \mathrm{H}(Y) = \mathrm{H}(Z) = \mathrm{h}\left(\tfrac{1}{2}\right) = 1 \text{ bit}.$$

Step 2 – Joint entropy of each pair. Pairwise independence $\mathrm{I}(X;Y) = 0$ means, by definition:

$$\mathrm{H}(X,Y) \;=\; \mathrm{H}(X) + \mathrm{H}(Y) - \underbrace{\mathrm{I}(X;Y)}_{=0} \;=\; \mathrm{H}(X) + \mathrm{H}(Y) \;=\; 2 \text{ bits}.$$

Likewise $\mathrm{H}(X,Z) = \mathrm{H}(Y,Z) = 2$ bits.

Step 3 – Lower bound on $\mathrm{H}(X,Y,Z)$. By the *chain rule for entropy*:

$$\mathrm{H}(X,Y,Z) \;=\; \mathrm{H}(X,Y) + \mathrm{H}(Z \mid X,Y) \;=\; 2 + \mathrm{H}(Z \mid X,Y).$$

Since $\mathrm{H}(Z \mid X,Y) \geq 0$ always,

$$\mathrm{H}(X,Y,Z) \;\geq\; 2 \text{ bits}.$$

Step 4 – Characterisation of equality. The lower bound is achieved ($\mathrm{H}(X,Y,Z) = 2$) *if and only if*

$$\mathrm{H}(Z \mid X,Y) = 0 \iff Z \text{ is a deterministic function of } (X,Y).$$

We must verify there *exists* such a function satisfying all pairwise independence constraints — this is done in part (b).

$$\boxed{\min \mathrm{H}(X, Y, Z) = 2 \text{ bits.}}$$

(b) Achieving example: $Z = X \oplus Y$

Construction. Let X and Y be i.i.d. Bernoulli($\frac{1}{2}$), and define $Z := X \oplus Y$ (XOR, equivalently addition mod 2).

Verification 1 – Z is Bernoulli($\frac{1}{2}$).

$$\mathbf{P}(Z = 0) = \mathbf{P}(X = Y) = \mathbf{P}(X = 0, Y = 0) + \mathbf{P}(X = 1, Y = 1) = \tfrac{1}{4} + \tfrac{1}{4} = \tfrac{1}{2}. \quad \checkmark$$

Verification 2 – Pairwise independence. Since X, Y are i.i.d., $\mathrm{I}(X; Y) = 0$ by assumption.

For $X \perp Z$: for any $(x, z) \in \{0, 1\}^2$, the unique value $y = z \oplus x$ satisfies $(X = x) \cap (Z = z) = (X = x) \cap (Y = z \oplus x)$, so

$$\mathbf{P}(X = x,\, Z = z) = \mathbf{P}(X = x,\, Y = z \oplus x) = \underbrace{\mathbf{P}(X = x)}_{1/2} \cdot \underbrace{\mathbf{P}(Y = z \oplus x)}_{1/2} = \frac{1}{4} = \mathbf{P}(X = x) \cdot \mathbf{P}(Z = z). \quad \checkmark$$

For $Y \perp Z$: since $Z = X \oplus Y = Y \oplus X$ (XOR is commutative), for any $(y, z) \in \{0, 1\}^2$ the unique value $x = z \oplus y$ gives $(Y = y) \cap (Z = z) = (X = z \oplus y) \cap (Y = y)$, so

$$\mathbf{P}(Y = y,\, Z = z) = \mathbf{P}(X = z \oplus y,\, Y = y) = \underbrace{\mathbf{P}(X = z \oplus y)}_{1/2} \cdot \underbrace{\mathbf{P}(Y = y)}_{1/2} = \frac{1}{4} = \mathbf{P}(Y = y) \cdot \mathbf{P}(Z = z). \quad \checkmark$$

Verification 3 – Entropy. Z is fully determined by (X, Y), so $\mathrm{H}(Z \mid X, Y) = 0$, and:

$$\mathrm{H}(X, Y, Z) = \mathrm{H}(X, Y) + \mathrm{H}(Z \mid X, Y) = 2 + 0 = 2 \text{ bits.} \quad \checkmark$$

The lower bound of Step 4 is therefore *tight*.

Non-mutual independence. Although X, Y, Z are pairwise independent, they are **not** mutually independent: $\mathrm{H}(X, Y, Z) = 2 < 3 = \mathrm{H}(X) + \mathrm{H}(Y) + \mathrm{H}(Z)$. Equivalently, knowing both X and Y completely determines Z — a clear dependence. This is a classical counterexample showing that *pairwise independence does not imply mutual independence.*

5. Problem 3.3 – Piece of Cake

Problem Statement

A cake is sliced roughly in half; the largest piece is always chosen and the other discarded. Each random cut creates pieces in proportions

$$P = \begin{cases} \left(\frac{2}{3}, \frac{1}{3}\right) & \text{with probability } \frac{3}{4}, \\ \left(\frac{3}{5}, \frac{2}{5}\right) & \text{with probability } \frac{1}{4}. \end{cases}$$

How large, *to first order in the exponent*, is the piece of cake after n cuts?

Background

Theorem 5.1 (Weak Law of Large Numbers)**.** *Let $W_1, W_2, \ldots$ be i.i.d. with $\mathbf{E}[|W_1|] < \infty$. Then $\frac{1}{n}\sum_{k=1}^{n} W_k \xrightarrow{\mathbf{P}} \mathbf{E}[W_1]$ as $n \to \infty$.*

Definition 5.2 (First-order exponent equality)**.** We write S_n $expeq 2^{nR}$ to mean $\frac{1}{n}\log_2 S_n \to R$ in probability as $n \to \infty$.

Solution

Step 1 – Model. At each cut i, the surviving (largest) piece is multiplied by

$$M_i = \begin{cases} \frac{2}{3} & \text{with probability } \frac{3}{4}, \\ \frac{3}{5} & \text{with probability } \frac{1}{4}, \end{cases}$$

with $M_1, M_2, \ldots$ i.i.d. The cake size after n cuts (starting from size 1) is

$$S_n = \prod_{i=1}^{n} M_i.$$

Step 2 – Take logarithms.

$$\frac{1}{n}\log_2 S_n = \frac{1}{n}\sum_{i=1}^{n} \log_2 M_i.$$

Step 3 – Verify the LLN condition. Since $M_i \in \{\frac{2}{3}, \frac{3}{5}\}$ are bounded and positive, $|\log_2 M_i|$ is bounded, so $\mathbf{E}[|\log_2 M_1|] < \infty$ trivially. By the WLLN:

$$\frac{1}{n}\log_2 S_n \xrightarrow{\mathbf{P}} \mathbf{E}[\log_2 M] \quad \text{as } n \to \infty.$$

Step 4 – Compute $\mathbf{E}[\log_2 M]$.

$$\begin{aligned} \mathbf{E}[\log_2 M] &= \frac{3}{4}\log_2\frac{2}{3} + \frac{1}{4}\log_2\frac{3}{5} \\ &= \frac{3}{4}\Big(\log_2 2 - \log_2 3\Big) + \frac{1}{4}\Big(\log_2 3 - \log_2 5\Big) \\ &= \frac{3}{4} - \frac{3}{4}\log_2 3 + \frac{1}{4}\log_2 3 - \frac{1}{4}\log_2 5 \\ &= \frac{3}{4} - \frac{1}{2}\log_2 3 - \frac{1}{4}\log_2 5. \end{aligned}$$

Numerical evaluation: $\log_2 3 \approx 1.58496$, $\log_2 5 \approx 2.32193$.

$$\mathbf{E}[\log_2 M] \approx 0.750 - \tfrac{1}{2}(1.585) - \tfrac{1}{4}(2.322) = 0.750 - 0.793 - 0.581 = -0.624.$$

Step 5 – Size to first order in the exponent. By the definition of *expeq*:

$$S_n \mathit{expeq}\ 2^{n\,\mathbf{E}[\log_2 M]} = \left(\frac{2}{3}\right)^{\frac{3n}{4}} \cdot \left(\frac{3}{5}\right)^{\frac{n}{4}}.$$

Step 6 – Closed-form exponent.

$$\left(\frac{2}{3}\right)^{\frac{3n}{4}} \cdot \left(\frac{3}{5}\right)^{\frac{n}{4}} = \left[\left(\frac{2}{3}\right)^3 \cdot \frac{3}{5}\right]^{n/4} = \left[\frac{8}{27} \cdot \frac{3}{5}\right]^{n/4} = \left(\frac{8}{45}\right)^{n/4}.$$

Clarification on the exponent notation. The exponent $n/4$ applies to the *entire* fraction $(8/45)$, not to 45 alone:

$$\left(\frac{8}{45}\right)^{n/4} = \frac{8^{n/4}}{45^{n/4}} \neq \frac{8}{45^{n/4}}.$$

Numerical check at $n = 8$: $\left(\frac{8}{45}\right)^2 = \frac{64}{2025} \approx 0.0316$, whereas $8/45^2 = 8/2025 \approx 0.00395$ — a factor of 8 difference.

$$\boxed{S_n \mathit{expeq}\ \left(\frac{2}{3}\right)^{\frac{3n}{4}} \cdot \left(\frac{3}{5}\right)^{\frac{n}{4}} = \left(\frac{8}{45}\right)^{n/4} \approx 2^{-0.624\,n}.}$$

Interpretation. The factor $2^{-0.624} \approx 0.649$ is the *geometric mean* of the cut factors:

$$2^{\mathbf{E}[\log_2 M]} = \left(\frac{2}{3}\right)^{3/4} \cdot \left(\frac{3}{5}\right)^{1/4} \approx 0.649.$$

This is *not* the value of any single cut (individual cuts give $2/3 \approx 0.667$ or $3/5 = 0.600$), but rather the *log-average* reduction rate per cut. In other words, $S_n \approx (0.649)^n$ in the exponential sense: each of the n cuts contributes an average of ≈ -0.624 bits to $\log_2 S_n$. The cake shrinks to zero exponentially fast.

Almost-sure convergence. Since M_i are bounded, $\mathbf{E}[(\log_2 M_1)^2] < \infty$ as well, so in fact the Strong Law of Large Numbers gives $\frac{1}{n}\log_2 S_n \xrightarrow{\text{a.s.}} \mathbf{E}[\log_2 M]$, a stronger statement than convergence in probability.